Digital Activity Phenotypes in Depression: PCA and Hierarchical Clustering of Wrist Actigraphy

Abstract

Wearable activity data could offer objective, continuous "digital proxies" of behaviour in depression, but it is unclear whether such data reveal discrete behavioural subgroups or a continuous gradient. This project analysed minute-level wrist actigraphy from 55 adults (23 with depression, 32 controls) in the public Depression dataset. Ten activity and circadian-rhythm features were engineered per person, reduced with principal component analysis (PCA), and grouped with Ward hierarchical clustering.

Two components explained 71% of the variance: PC1 (45%) captured reduced, fragmented activity and PC2 (26%) captured night-time activity versus rhythm strength. Silhouette analysis favoured two clusters (0.49), separating a small group (n = 8, 75% depressed) with markedly low and irregular activity. However, this partition was only moderately stable under bootstrap resampling, changed under other linkage methods, and was weakly associated with diagnosis (Fisher $p = 0.057$). The continuous PC1 score separated depressed participants from controls better (AUC 0.72, $p = 0.007$), including after excluding inpatients. No component was related to depression severity (MADRS).

The results suggest that activity patterns in depression form a continuum rather than discrete subtypes, and that dimensional digital proxies are more informative than cluster membership.

Introduction

Major depressive disorder (MDD) is commonly assessed with clinical interviews and questionnaires. These are valuable but infrequent, partly subjective, and mix many symptoms into one score. Wearable sensors offer a complementary view: they record behaviour continuously in daily life and can yield objective digital proxies of constructs such as motivation, energy and sleep–wake regulation.

Reduced activity and disturbed circadian rhythms are well-documented features of depression. Psychomotor retardation, loss of motivation (anhedonia and apathy) and disrupted sleep can all leave traces in actigraphy. Non-parametric circadian measures, such as interdaily stability and intradaily variability, summarise how regular and fragmented a person's rest–activity pattern is.

Depression is also heterogeneous: people with the same diagnosis can show very different symptom profiles. A natural question is therefore whether activity data reveal distinct behavioural subgroups. Unsupervised methods such as PCA and hierarchical clustering suit this exploratory question. PCA summarises many correlated features into a few interpretable dimensions, and hierarchical clustering shows how participants group at every level of similarity. A key risk, however, is that clustering always produces groups, even when none exist. This project therefore places as much weight on evaluating the clusters as on finding them.

Research Question

The project asked whether wearable activity data reveal natural behavioural subgroups in depression, and whether these relate to clinical status. This was broken into three questions:

1. **Structure:** Which underlying dimensions summarise daily activity and circadian-rhythm features, and do participants form distinct clusters along them?
2. **Validity:** Are any clusters robust to resampling and to analytical choices such as the linkage method and the use of PCA?
3. **Clinical relevance:** Do the clusters or the underlying dimensions relate to depression diagnosis and severity, beyond confounds such as age, sex and inpatient status?

Data

The analysis used the public Depression dataset (Garcia-Ceja et al., 2018), collected in Norway. Each participant wore an Actiwatch on the right wrist, which recorded an activity count every minute.

Characteristic	Depressed	Controls
Participants	23	32
Female / male	10 / 13	20 / 12
Diagnosis	15 unipolar, 7 bipolar II, 1 bipolar I	—
Setting	18 outpatients, 5 inpatients	—
MADRS at start	mean 22.7 (range 13–29)	—

Age was recorded in 5-year bands; band midpoints were used as numeric age. To avoid bias from partially recorded days, only complete 24-hour days (1,440 minutes) were kept, up to the number of days reported per participant. This gave a median of 13 days per person (range 5–20). The MADRS (Montgomery–Åsberg Depression Rating Scale) score at the start of recording served as the severity measure.

Methodology

The analysis followed five steps, all implemented in Python (pandas, scikit-learn, SciPy) with a fixed random seed for reproducibility.

1. Feature engineering

Ten digital proxies were computed per participant, chosen to capture activity level, regularity and fragmentation.

Feature	What it measures
Mean activity	Mean of log(1 + count) over all minutes
% inactive	Share of minutes with zero activity
Daily variability (CV)	Coefficient of variation of daily total activity
Night share	Share of total activity between 00:00 and 06:00
Day fragmentation	Probability that an active minute is followed by rest (08:00–22:00)

Interdaily stability (IS)	How consistent the 24-hour pattern is across days (0–1)
Intradaily variability (IV)	How fragmented the rhythm is hour to hour (higher = more fragmented)
Relative amplitude (RA)	Contrast between most and least active periods (0–1)
M10	Activity in the most active 10 hours (log)
L5	Activity in the least active 5 hours (log)

IS, IV, RA, M10 and L5 are standard non-parametric circadian measures (van Someren et al., 1999), computed from hourly means.

2. Principal component analysis

Features were standardised (z-scores) so that no feature dominated because of its scale. PCA was then applied. Rather than relying on a single rule, the number of components was judged by three criteria: 80% cumulative explained variance, the Kaiser criterion (eigenvalue > 1), and Horn's parallel analysis, which keeps components whose eigenvalues exceed the 95th percentile of those from 500 random datasets of the same size. Components were interpreted through their loadings.

3. Hierarchical clustering

Agglomerative clustering with Ward linkage was applied to the retained PC scores. Ward linkage merges the pair of clusters that least increases within-cluster variance, which suits compact groups in a Euclidean PC space. The number of clusters k was chosen by the mean silhouette width for k = 2 to 6, alongside visual inspection of the dendrogram. The cophenetic correlation measured how faithfully the dendrogram preserved the original distances.

4. Internal validation

- **Bootstrap stability:** the full pipeline (standardisation, PCA, clustering) was refitted on 500 bootstrap samples. For each original cluster, the mean Jaccard similarity with its best-matching bootstrap cluster was recorded (Hennig, 2007). Values above 0.75 are usually considered stable and below 0.5 unstable.
- **Sensitivity to analytical choices:** the partition was compared, using the adjusted Rand index (ARI), with average- and complete-linkage solutions and with Ward clustering on all standardised features without PCA.

5. External validation

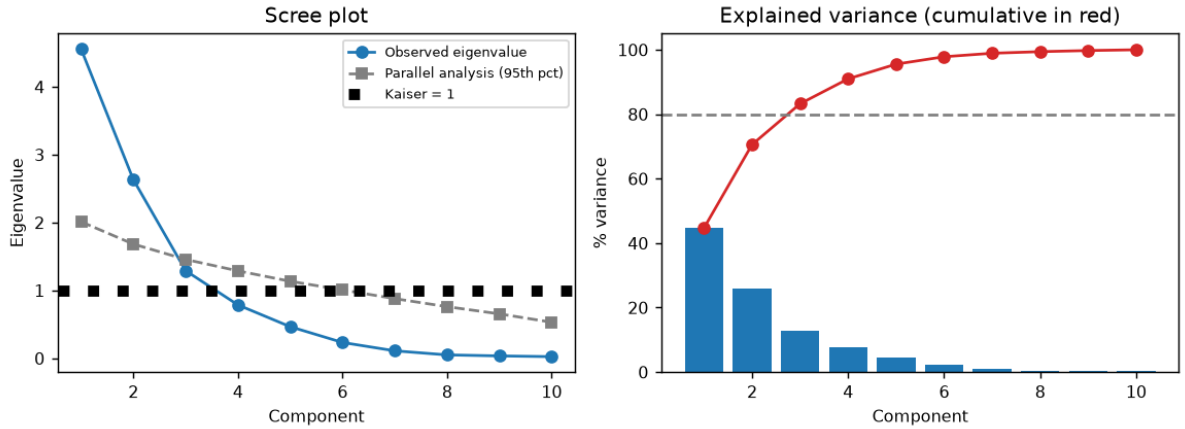
- Agreement between clusters and diagnosis (ARI, Fisher's exact test).
- Differences in PC scores between depressed participants and controls (Mann–Whitney U, AUC).
- Spearman correlations between PC scores and MADRS within the depressed group.
- Confound checks: age (Kruskal–Wallis), sex (chi-squared) and inpatient status, including a re-analysis restricted to outpatients and controls.

Results

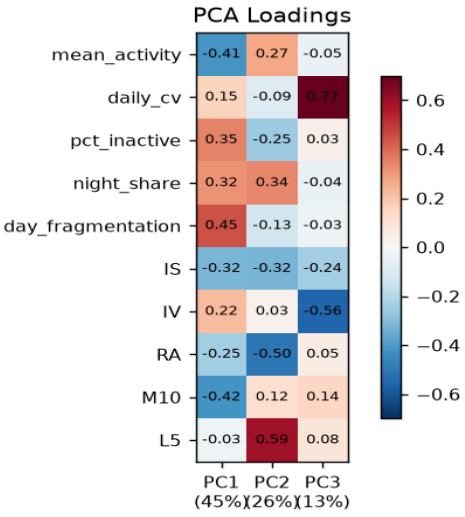
The data showed two clear activity dimensions, but only weak evidence for distinct subgroups. The continuous first component was more clinically informative than the clusters.

Principal components

The first three components explained 83% of the variance (PC1 45%, PC2 26%, PC3 13%). The 80% rule and the Kaiser criterion both suggested three components, while parallel analysis supported two. Three components were retained for clustering, as the third still carried meaningful variance; the first two were the focus of interpretation.

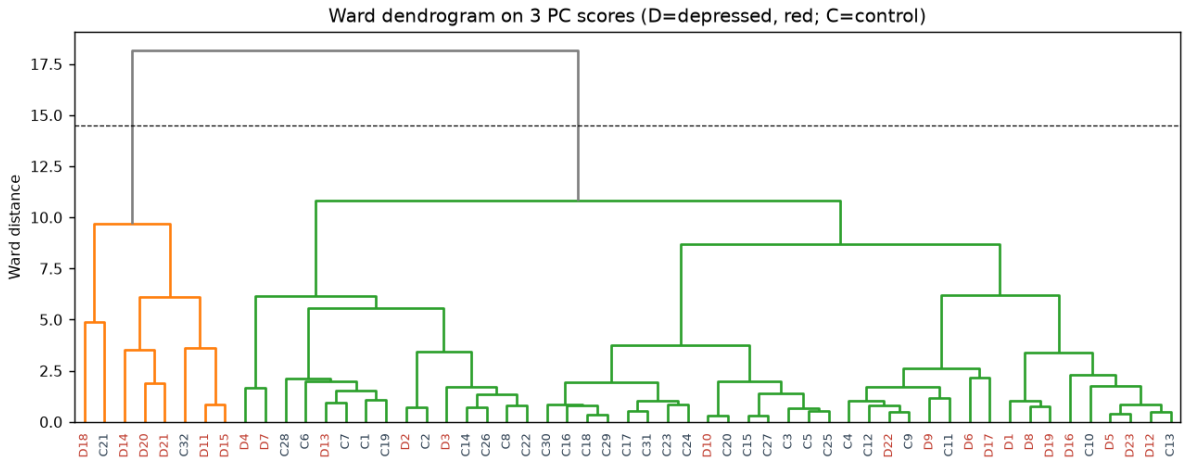


- **PC1 – reduced, fragmented activity.** High scores meant fragmented daytime activity (loading 0.45), more inactivity (0.35) and more night-time activity (0.32), with low mean activity (−0.41), low peak activity M10 (−0.42) and lower stability IS (−0.32).
- **PC2 – night-time activity versus rhythm strength.** High scores meant more activity in the quietest hours L5 (0.59) and at night (0.34), with lower relative amplitude (−0.50) and stability (−0.32).
- **PC3 – day-to-day variability.** Dominated by daily variability (0.77) and, inversely, by within-day fragmentation IV (−0.56).



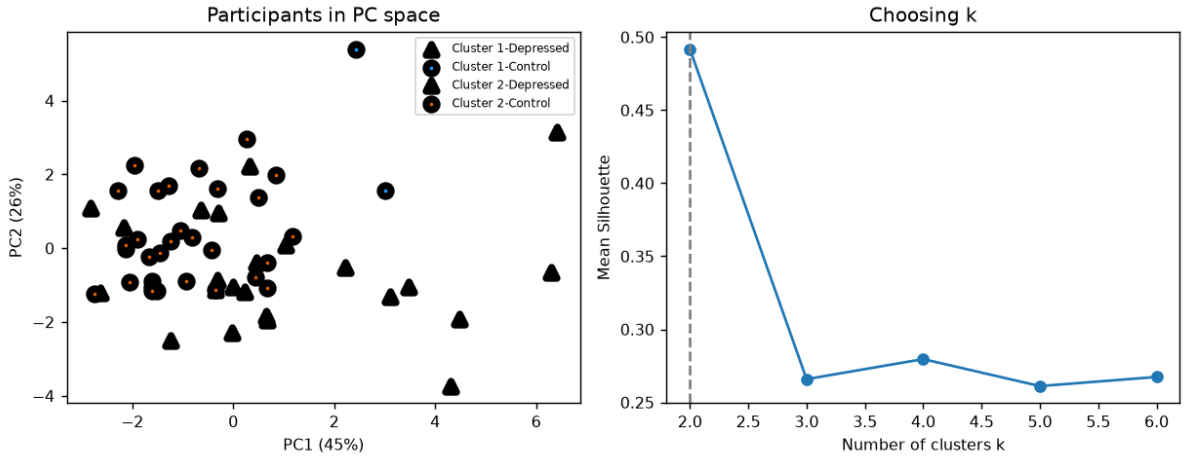
Clusters

Ward clustering on the three PC scores gave the highest silhouette at k = 2 (0.49); all other solutions scored 0.28 or below. The cophenetic correlation of 0.78 indicated that the dendrogram represented the distances reasonably well.

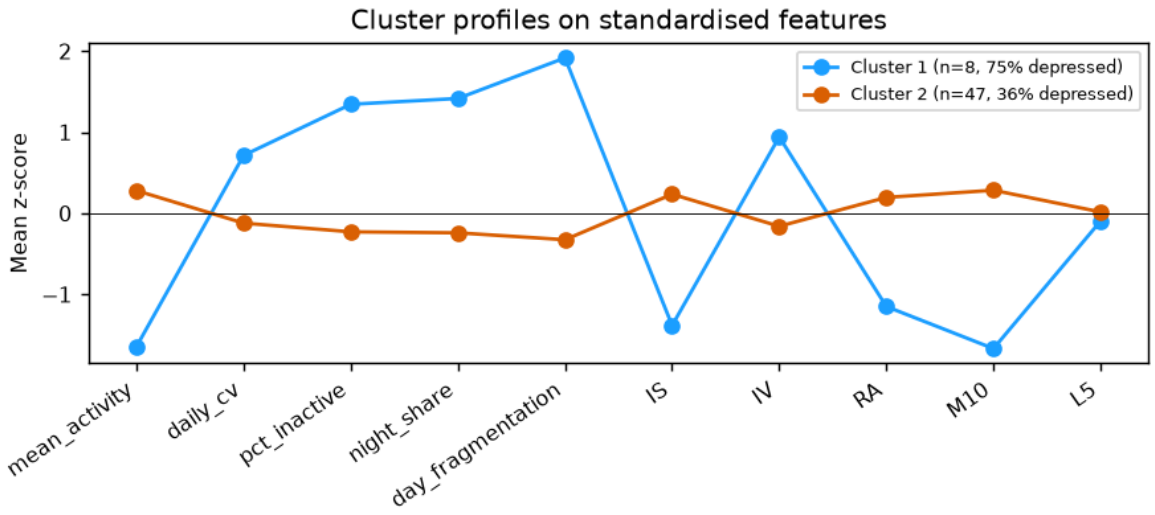


The two clusters were very unequal in size. The small cluster sat at the high end of PC1, rather than forming a separate cloud.

	Cluster 1	**Cluster 2**
Participants	8	47
Depressed	6 (75%)	17 (36%)
Mean activity (log)	2.25	3.46
% inactive minutes	52%	32%
Night share of activity	14%	5%
Interdaily stability	0.27	0.46
Relative amplitude	0.70	0.86
Mean age (years)	44.5	39.7



Cluster 1 was characterised by low, fragmented and irregular activity, with a flattened rest–activity rhythm.



Robustness

Check	Result	Interpretation
Bootstrap Jaccard, cluster 1	0.71	Moderately stable, below the 0.75 threshold
Bootstrap Jaccard, cluster 2	0.90	Stable
ARI vs average and complete linkage	0.31	Partition depends on the linkage method
ARI vs Ward without PCA	0.80	PCA step did not drive the result

Clinical relevance

- **Diagnosis and clusters:** depressed participants were over-represented in cluster 1, but the association was weak (ARI 0.08, Fisher $p = 0.057$). At $k = 3$ and $k = 4$ the extra clusters each contained a mix of patients and controls.
- **Diagnosis and components:** depressed participants scored higher on PC1 (Mann–Whitney $p = 0.007$, AUC 0.72) and lower on PC2 ($p = 0.010$; AUC 0.71 for controls scoring higher). Lower PC2 reflected quieter nights in the depressed group (median L5 2.86 vs 3.28).
- **Inpatient status:** inpatients did not differ significantly from outpatients on PC1 ($p = 0.29$). The PC1 difference remained when only outpatients were compared with controls ($p = 0.040$).
- **Severity:** no component correlated with MADRS within the depressed group (all $|\rho| \leq 0.11$, all $p > 0.6$). MADRS also did not differ between clusters (means 22.8 and 22.7).
- **Demographic confounds:** clusters did not differ by age ($p = 0.30$) or sex ($p = 1.0$), and age was not related to PC1 ($\rho = 0.15$, $p = 0.27$).

Conclusion

Activity patterns in this sample formed a continuum rather than discrete subtypes. Answering the three research questions:

1. **Structure:** two interpretable dimensions summarised the data well: reduced, fragmented activity (PC1) and night-time activity versus rhythm strength (PC2). The only well-supported cluster solution split off a small tail of participants at the high end of PC1, rather than revealing separate groups.
2. **Validity:** this tail cluster was only moderately stable and changed with the linkage method, so it should not be treated as a robust subtype. The result did not depend on the PCA step.
3. **Clinical relevance:** the continuous PC1 score distinguished depressed participants from controls (AUC 0.72), and this was not explained by hospitalisation, age or sex. Neither the clusters nor the components tracked MADRS severity.

The main lesson is methodological. Clustering always returns groups, so its output must be tested for stability and external validity before it is interpreted. Here, those tests pointed towards a dimensional proxy as the more useful summary.

The lack of association with MADRS has two plausible explanations. First, the severity range in this sample was narrow (13–29). Second, a single interview total mixes mood, cognitive and somatic symptoms, while activity may relate more specifically to constructs such as motivation, energy or psychomotor change.

References

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